

TAHA SHAHEEN

Tempe, AZ | (602) 565-7265 | taha.shaheen@asu.edu | linkedin.com/in/taha-shaheen | tahashaheen.github.io

SUMMARY

Ph.D. student in Computer Science investigating how a mismatch between a human's understanding of domain dynamics affects the quality of their feedback, and how this in turn impacts an agent's reward learning. My work bridges human behavior, robotics, and adaptive human-AI collaboration, with prior research on preventing misuse of teleoperated robots and developing low-cost socially assistive systems.

Availability: Seeking a full-time, 12-week Summer research internship beginning May 2026.

EDUCATION

Ph.D. Computer Science Arizona State University, USA <i>Jumpstart Research Grant, Teaching Excellence Award (2025)</i>	Expected May 2028 4.00/4.00 GPA
M.S. Social Informatics Kyoto University, Japan <i>MEXT Research Scholarship (Japanese Gov.), Distinguished Master's Thesis Award</i>	April 2023 3.92/4.00 GPA
B.E. Electronic Engineering NED University of Engineering and Technology, Pakistan <i>Academic Rank 5 out of 148</i>	November 2017 3.70/4.00 GPA

RESEARCH AND ENGINEERING EXPERIENCE

Graduate Student Researcher <i>Cooperative Robotic Systems Lab, Arizona State University, USA</i>	August 2023 – Present
- Designed and currently executing a complex 16-condition study investigating how perturbed user beliefs impact feedback quality in reward learning for AI agents and robots. - Created a full-stack online platform (Python, Flask, JS) for a large-scale remote user study, successfully collecting over 4,500 unique trajectories from 180+ participants recruited and managed via SONA Systems. - Deployed an automated pipeline for data collection and processing, and am currently analyzing behavioral data using Python (Pandas, Numpy, Scikit-learn) and SPSS.	
Graduate Student Researcher <i>Human-Robot Interaction Lab, Kyoto University, Japan</i>	October 2020 – August 2023
- Devised a novel role-playing study to explore adversarial behaviors yielding key insights for future robotic ethics frameworks where 12 participants acted as victims and as malicious operators using a physical robot as well as in a crowd simulation. - Identified and cataloged 15 unique “low-moral” actions (e.g., privacy violation, harassment), and developed a framework of technical and social countermeasures to prevent them. - Authored the resulting taxonomy and prevention framework, culminating in a first-author publication in the journal ACM Transactions on Human-Robot Interaction.	
Research Assistant <i>National Center for Artificial Intelligence, NED University, Pakistan</i>	November 2018 – September 2020
- Engineered 2 socially assistive robots from concept to deployment for a target cost under \$200 USD, developing the full software/hardware stack in Unity (C#), Android (Java), and Arduino (C++). - Drove 3 major robot iterations by incorporating insights from a 3-month field study and targeted feedback from 15 key stakeholders (therapists and parents). - Mentored 12 undergraduate students across 3 capstone teams on robotics projects for autism therapy, leading all 3 teams to successfully develop and validate novel therapeutic robot prototypes.	

TECHNICAL SKILLS

Software & Tools: SPSS, SONA for Participant Management, Unity, Docker, Fusion 360, Android Studio
Programming Languages: Python, C++, Java, C#, R, Kotlin
Libraries & Frameworks: PyTorch, ROS 2, Pandas, NumPy, Scikit-learn, Flask, Matplotlib
Hardware & Robotics: Universal Robots UR5, Raspberry Pi, Arduino

NOTABLE PROJECTS

- LLM as a Model Optimizer in Model Predictive Control** 2025
- **Demonstrated** that an LLM can function as a real-time model optimizer for an MPC system, successfully adapting the controller parameters online to stabilize a dynamic inverted pendulum simulation.
 - **Engineered** a physical testbed for hardware validation by designing and 3D-printing a custom pendulum and integrating it with a UR5 robotic arm.
- Active Shadowing (ASD) for Implicit HRI Communication** 2025
- **Developed** the core robot motion control, precisely choreographing a UR5 arm's movements to seamlessly integrate with a predictive Augmented Reality shadow projected via a HoloLens.
 - **Co-designed** the 23-participant user that validated the system, demonstrating a statistically significant improvement in human-robot safety scores by over 50% compared to explicit communication.
- Office Reception Robot** 2020
- **Engineered** the robot's perception and interaction systems, integrating a Python-based computer vision module for facial recognition and Google Dialogflow for natural language understanding (NLU).
 - **Built** the full-stack architecture, including a Python back-end and 3 custom Android front-ends, to successfully automate 3 core office tasks (guest registration, package delivery, and meeting scheduling).
- Assistive Social Robot (Capstone Project)** 2017
- **Engineered** a social robot from the ground up, developing the full software stack using Unity (C#), Android (Java), and Arduino (C++) as a key member of a 4-person capstone team. The project's novelty and success was instrumental in securing a **\$26,000** national grant, funding its evolution into a formal research position.

PUBLICATIONS

Andrew Boateng, Prakhar Bhartiya, **Taha Shaheen**, Yu Zhang. 2025. "Active Shadowing (ASD): Manipulating Perception of Robotic Behaviors via Implicit Virtual Communication." arXiv preprint arXiv: arXiv:2407.01468

Winston Smith, Andrew Boateng, **Taha Shaheen**, and Yu Zhang. 2025. "Assigning Multi-Robot Tasks to Multitasking Robots." arXiv preprint arXiv:2506.15032.

Taha Shaheen, Dražen Brščić, and Takayuki Kanda. 2024. Investigation of Low-Moral Actions by Malicious Anonymous Operators of Avatar Robots. *ACM Transactions on Human-Robot Interaction*. doi.org/10.1145/3696466.

LEADERSHIP & VOLUNTEER EXPERIENCE

- Oral History Apprentice & Field Officer** 2016 – 2019
1947 Partition Archive
- Led a team of 3 volunteers and personally interviewed and filmed 83 witnesses of the 1947 Partition of British India, with work featured on BBC Radio and in Stanford University's library collection.